

A Rail-Based Personal Transit Framework for Next-Generation Urban and Intercity Transport

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Abstract

This paper proposes a rail-based personal transit system as a comprehensive replacement for conventional road traffic. The system employs small, autonomous electric trains carrying up to six passengers each, operating across a hierarchical network of urban and intercity rail circuits. Key design features include zone-based circuit partitioning, priority override protocols for emergency and public service vehicles, underground pedestrian crossings, and provisions for heavy freight transport. The proposed system eliminates the need for private vehicle ownership, parking infrastructure, and human drivers, while offering significantly improved safety, energy efficiency, and all-weather operational reliability. Integration with a tidal energy generation network — described in a companion paper — is proposed as the electricity supply source.

1. Introduction

Contemporary road-based transport systems present well-documented challenges including greenhouse gas emissions, traffic congestion, road accident fatalities, and extensive land use for parking and road infrastructure. Autonomous vehicle technology has been proposed as one mitigation strategy; however, it retains many of the fundamental inefficiencies of the private car paradigm. This paper instead proposes a fundamental restructuring of personal transport around a rail-based network of small, fully automated electric trains. The concept draws on the reliability and efficiency of existing rail systems while adapting them to the flexibility and accessibility demands of urban personal mobility.

2. System Architecture and Vehicle Design

The proposed system replaces private road vehicles with small rail-bound pods, each designed to carry up to six passengers. Larger travel groups would divide into sub-groups and board consecutive pods. Rail layouts consist of a primary travel lane supplemented by lateral sidings, which allow pods to decelerate and stop for passenger boarding and alighting without disrupting through traffic. A two-tier speed hierarchy is proposed: higher-speed intercity or arterial rail corridors connecting districts and cities, and lower-speed urban rail networks operating within populated centers. Passengers select their destination via a smartphone application or onboard interface, with routing and dispatch handled autonomously by the network control system.

3. Circuit Partitioning and Fleet Management

In large urban areas, the rail network is proposed to be divided into discrete geographic circuits, each operated by a dedicated sub-fleet of pods. This partitioning prevents pods from traveling excessive distances from their assigned service areas following passenger drop-offs, ensuring consistent availability throughout the network. Inter-circuit connections would exist but be reserved for fleet redistribution, maintenance movements, and the transport of heavy equipment. The operational logic is analogous to positional zoning in team sports, where units maintain area coverage while retaining the ability to operate outside their zone when operationally necessary. When a pod completes a journey near a circuit boundary, the network management system directs it to the nearest in-circuit station for reintegration into local service. A four-track configuration is recommended for high-demand corridors: two directional travel tracks and two dedicated stopping tracks, providing separation between through-running and boarding operations.

4. Priority Override and Emergency Service Integration

Emergency services — including ambulance, fire, and law enforcement — as well as designated public officials, would be granted priority access via a mobile application interface. Upon activation, the network management system would implement a corridor-clearing protocol: pods along the designated route would receive instructions to accelerate or yield, and at intersections, a configurable number of priority-route pods would be granted passage before standard-mode pods are permitted to proceed. This approach mirrors the signal preemption systems used

by modern light rail networks but extends the concept to dynamic, application-triggered full-corridor prioritization.

5. Heavy Freight and Equipment Transport

The transport of oversized or heavy equipment — such as construction machinery and excavators — is accommodated through the use of flatbed rail carriers operating on the standard network. In cases where load dimensions or weight require it, multiple parallel tracks may be utilized simultaneously. For exceptional transport operations, normally bidirectional track segments may be temporarily converted to unidirectional operation. During such periods, pods ordinarily assigned to the affected track are rerouted to adjacent circuits via the inter-circuit connection infrastructure, with the network management system coordinating reallocation to maintain service continuity.

6. Pedestrian Infrastructure

At-grade pedestrian crossings of rail corridors are replaced by standardized underground passages accessed via elevators at regular intervals along the network. In the event of elevator malfunction, units are designed to permit manual door operation, with internal ladders providing safe egress. A fault reporting interface integrated into the elevator system enables passengers to notify the responsible operator directly via mobile device, facilitating rapid response.

7. Key Advantages Over Conventional Road Transport

The proposed system offers the following improvements over the current road transport paradigm:

1. **Environmental impact:** The system operates entirely on electricity, eliminating direct vehicular emissions. When powered by the tidal energy generation network described in the companion paper, the system would have a near-zero operational carbon footprint.
2. **Land use:** The elimination of private vehicle ownership removes the requirement for parking facilities, freeing substantial urban land area for alternative uses.
3. **Automation:** Fully autonomous operation removes the need for driving licenses and human operators, reducing labor costs and eliminating human error as a cause of accidents.
4. **Passenger convenience:** Destination selection via smartphone provides a user experience comparable to ride-hailing services, without the inefficiencies of individually owned vehicles.
5. **Safety:** Fixed-track operation with automated control significantly reduces collision risk compared to road traffic. The deterministic nature of rail-based routing enables far more reliable hazard prevention than road-based autonomous systems.
6. **All-weather reliability:** Unlike road vehicles — which are subject to reduced performance in fog, ice, snow, and high winds — a rail-based system with enclosed or weather-protected infrastructure can maintain consistent service under adverse meteorological conditions.

8. Conclusion

The rail-based personal transit system proposed in this paper represents a substantive alternative to the private automobile as the primary mode of personal transport. By combining autonomous pod operation, hierarchical circuit management, emergency priority protocols, and integration with renewable energy infrastructure, the system addresses the core deficiencies of current road transport in terms of safety, emissions, land use, and operational reliability. Future work should address detailed civil engineering requirements for network deployment, economic modeling of transition costs, passenger capacity analysis under varying demand conditions, and the design of network management algorithms for real-time fleet optimization.

Keywords: *rail transport, autonomous transit, urban mobility, personal rapid transit, electric vehicles, smart infrastructure, sustainable transport, fleet management*